ASSIGNMENT-10.5

Lab 10 – Code Review and Quality: Using AI to Improve Code

Quality and Readability

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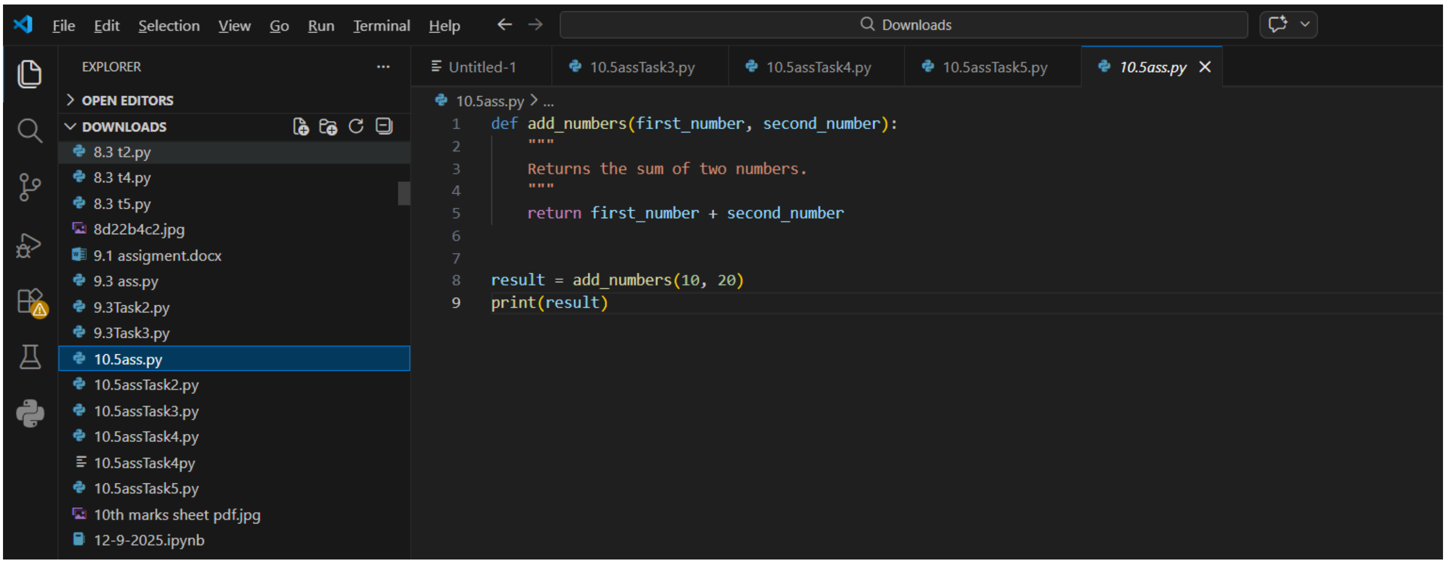
Task 1: Improving Variable & Function Names Original Code

def f(a, b):   return

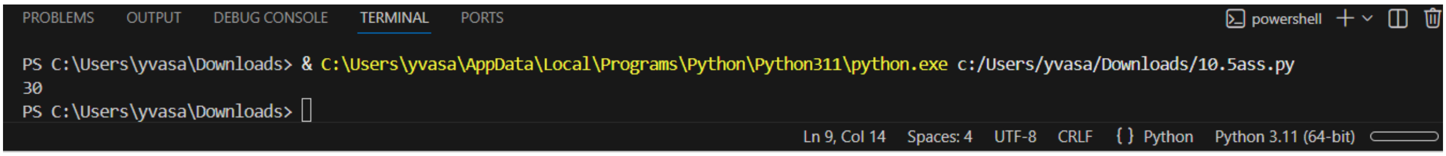
a + b print(f(10,

20))

CODE:



OUTPUT



Task Description #2 – Missing Error Handling

Task: Use AI to add proper error handling.

Sample Input Code: def

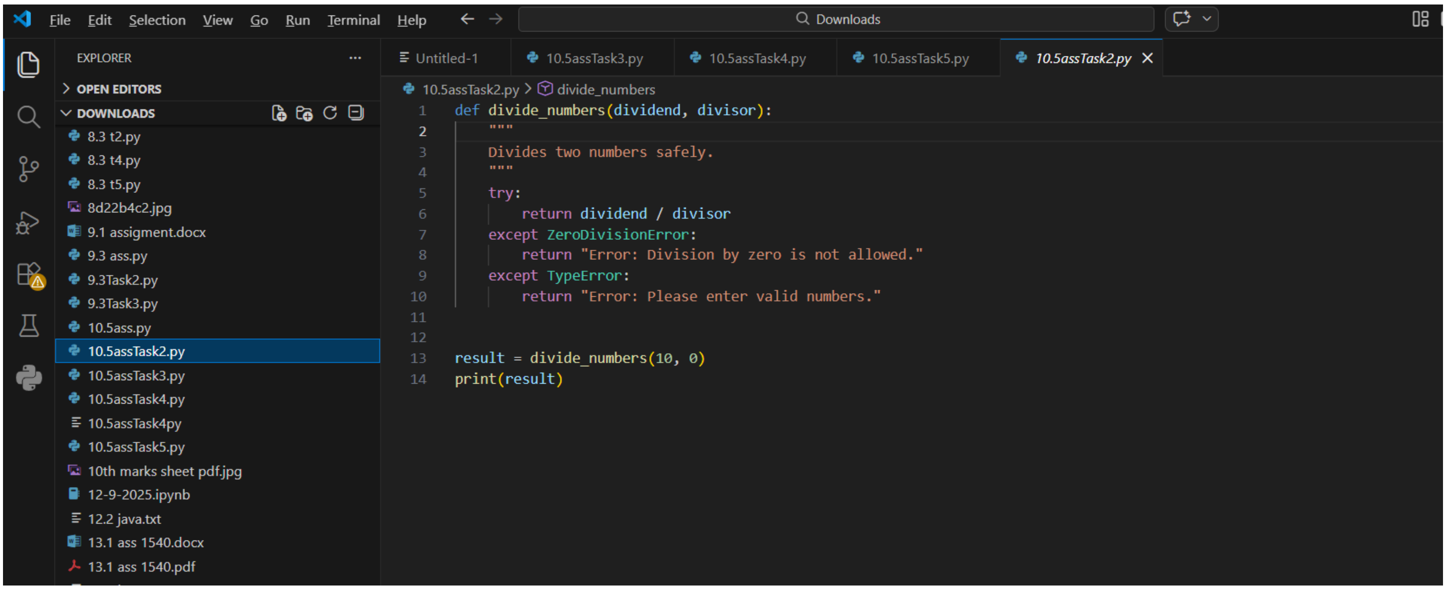
divide(a, b): return a / b

print(divide(10, 0))

Expected Output:

• Code with exception handling and clear error messages

CODE :



OUTPUT:

Task Description #3: Student Marks Processing System

The following program calculates total, average, and grade of a student, but it has poor readability, style issues, and no error handling.

marks=[78,85,90,66,88] t=0

for i in marks:

t=t+i

a=t/len(marks) if

a>=90: print("A")

elif a>=75:

print("B") elif

a>=60: print("C")

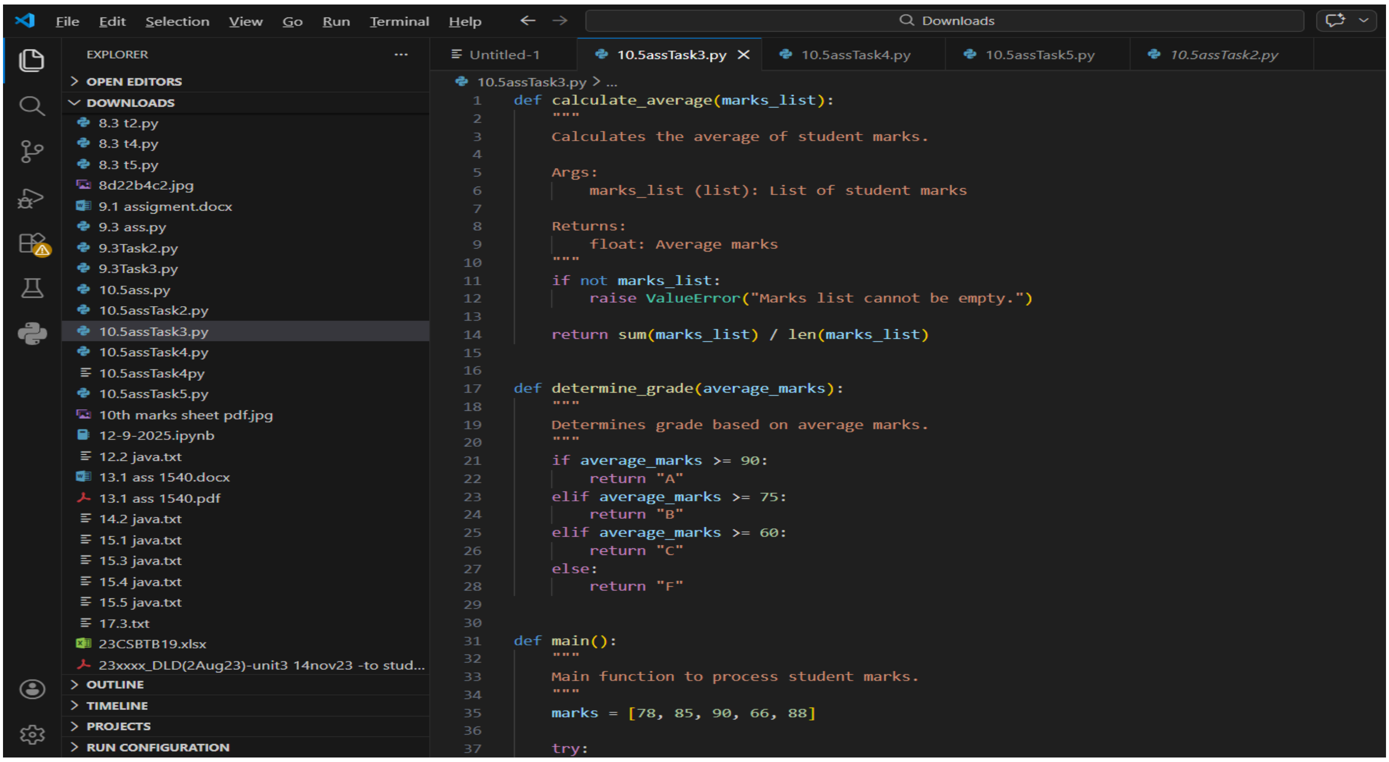
else:

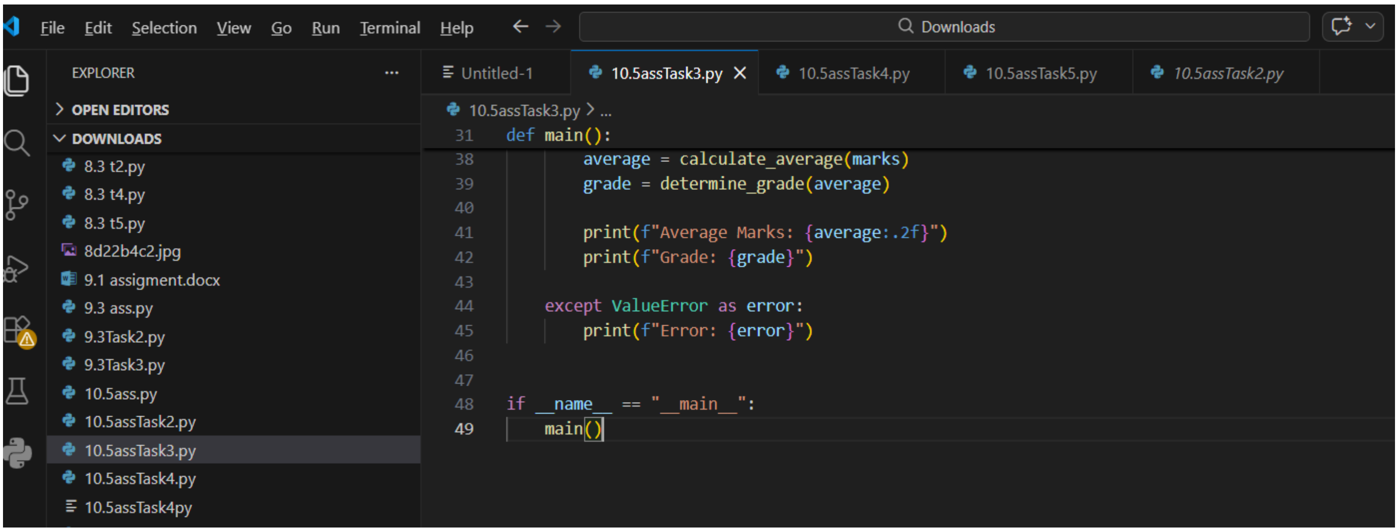
print("F")

Task:

- Use AI to refactor the code to follow PEP 8 standards.
- Add meaningful variable names, functions, and comments.
- Add basic input validation and documentation.

CODE :





OUTPUT:

```
PS C:\Users\yvasa\Downloads> & C:\Users\yvasa\AppData\Local\Programs\Python\Python311\python.exe c:/Users/yvasa/Downloads/10.5assTask3.py
Average Marks: 81.40
Grade: B
PS C:\Users\yvasa\Downloads>
```

Task Description #4: Use AI to add docstrings and inline comments to the

following function. def factorial(n):

result = 1 for i in

range(1,n+1):

result *= i

return result

CODE :

```python
def factorial(number):
    """
    Calculates the factorial of a given number.
    """
    if number < 0:
        raise ValueError("Negative numbers not allowed.")

    result = 1

    for i in range(1, number + 1):
        result *= i

    return result


if __name__ == "__main__":
    number = int(input("Enter a number: "))
    print("Factorial:", factorial(number))
```

OUTPUT:

```
PS C:\Users\yvasa\Downloads> & C:\Users\yvasa\AppData\Local\Programs\Python\Python311\python.exe c:/Users/yvasa/Downloads/10.5assTask4.py
Enter a number: 5
Factorial: 120
PS C:\Users\yvasa\Downloads>
```

Task Description #5: Password Validation System (Enhanced) The following

Python program validates a password using only a minimum length check,

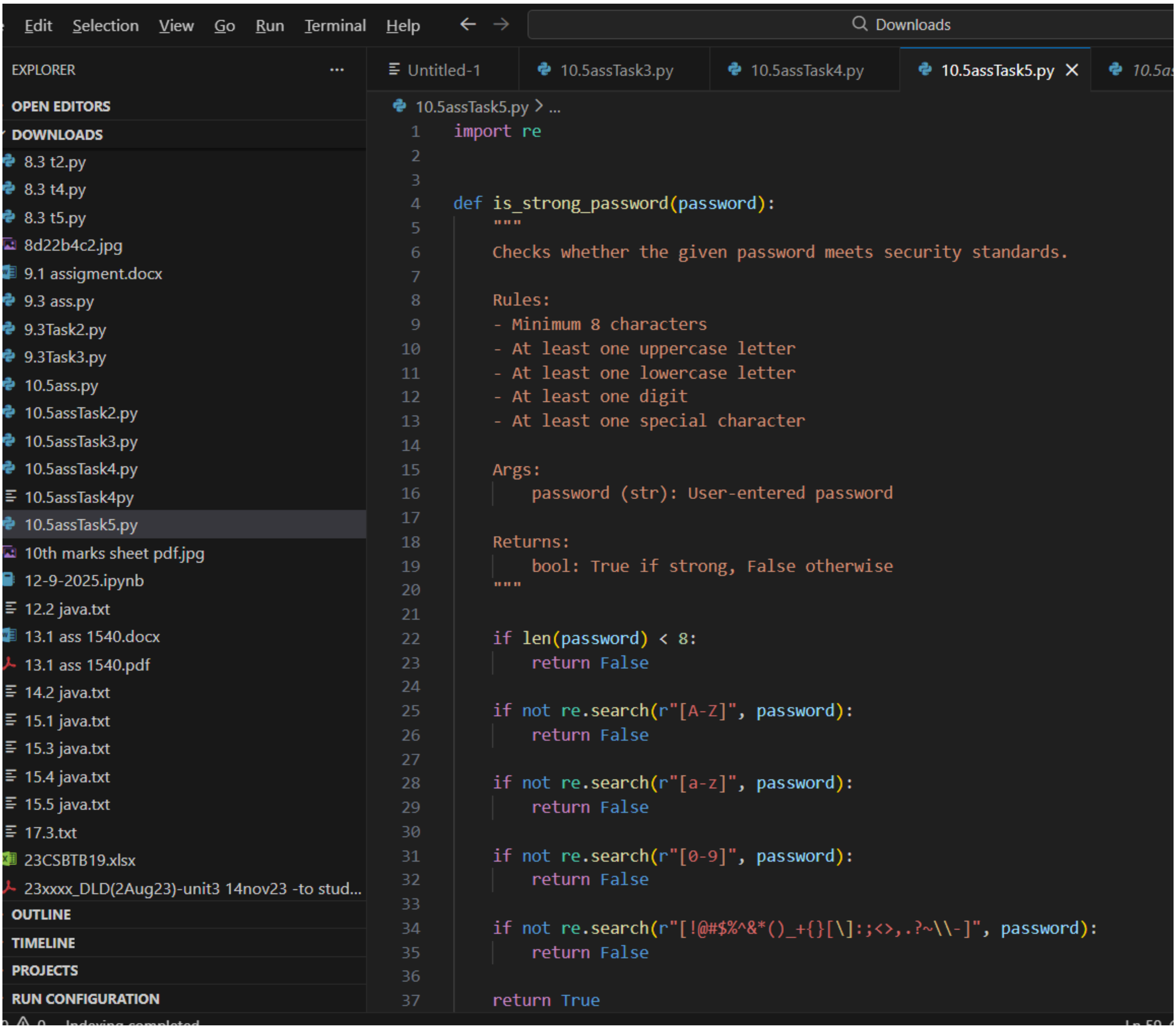
which is insufficient for real-world security requirements.

pwd = input("Enter password: ") if

len(pwd) >= 8: print("Strong") else:

print("Weak")

CODE :




```python
import re


def is_strong_password(password):
    """
    Checks whether the given password meets security standards.

    Rules:
    - Minimum 8 characters
    - At least one uppercase letter
    - At least one lowercase letter
    - At least one digit
    - At least one special character

    Args:
        password (str): User-entered password

    Returns:
        bool: True if strong, False otherwise
    """

    if len(password) < 8:
        return False

    if not re.search(r"[A-Z]", password):
        return False

    if not re.search(r"[a-z]", password):
        return False

    if not re.search(r"[0-9]", password):
        return False

    if not re.search(r"[!@#$%^&*()_+{}[\]:;<>,.?~\\-]", password):
        return False

    return True
```

```python
def main():
    """
    Main function for password validation.
    """
    user_password = input("Enter password: ")

    if is_strong_password(user_password):
        print("Password is Strong ✅")
    else:
        print("Password is Weak ❌")
        print("It must contain:")
        print("- At least 8 characters")
        print("- One uppercase letter")
        print("- One lowercase letter")
        print("- One digit")
        print("- One special character")


if __name__ == "__main__":
    main()
```

OUTPUT

```
Factorial: 120
PS C:\Users\yvasa\Downloads> & C:\Users\yvasa\AppData\Local\Programs\Python\Python311\python.exe c:/Users/yvasa/Downloads/10.5assTask5.py
Enter password: sara123
Password is Weak ❌
It must contain:
- At least 8 characters
- One uppercase letter
- One lowercase letter
- One digit
- One special character
PS C:\Users\yvasa\Downloads> & C:\Users\yvasa\AppData\Local\Programs\Python\Python311\python.exe c:/Users/yvasa/Downloads/10.5assTask5.py
Enter password: Sahasra@143
Password is Strong ✅
PS C:\Users\yvasa\Downloads>
```